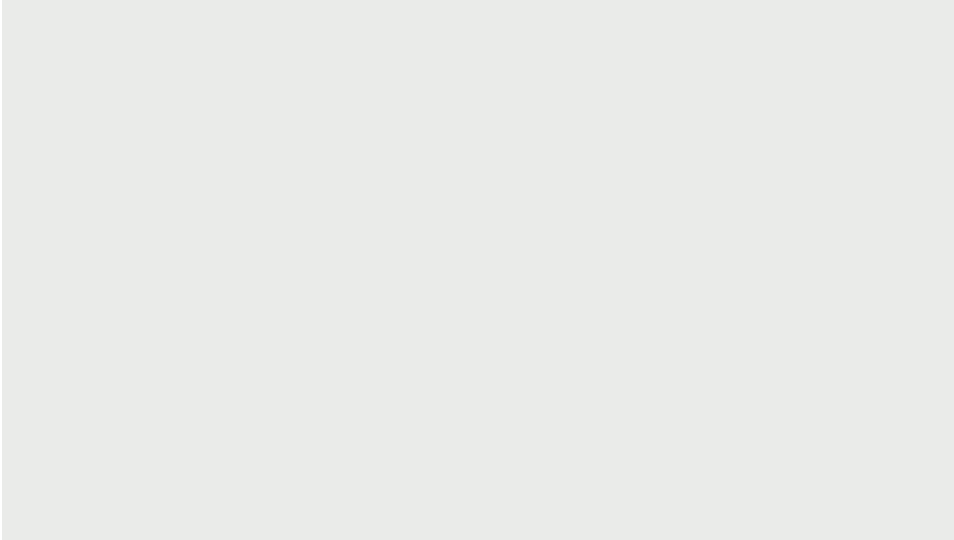
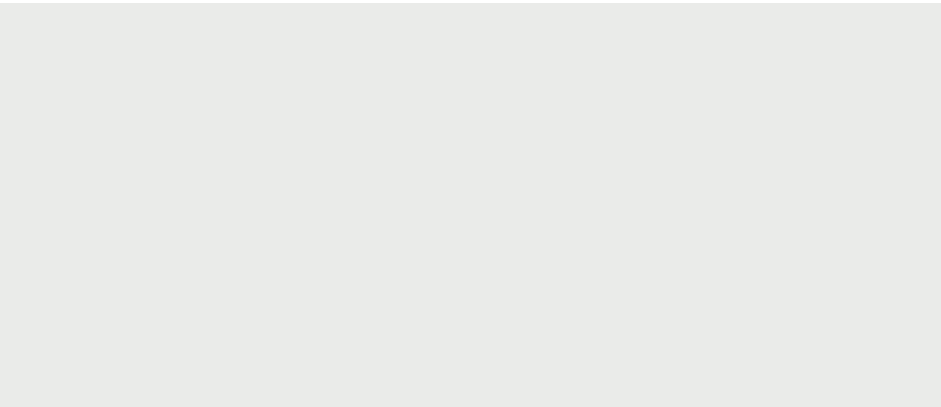




CONTRASTIVE LEARNING

By Hiba Shahid, Roozain Zehra and Amnah Qureshi



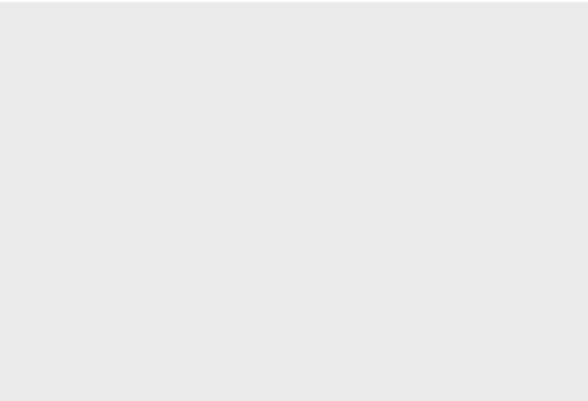
21 Nov, 2025



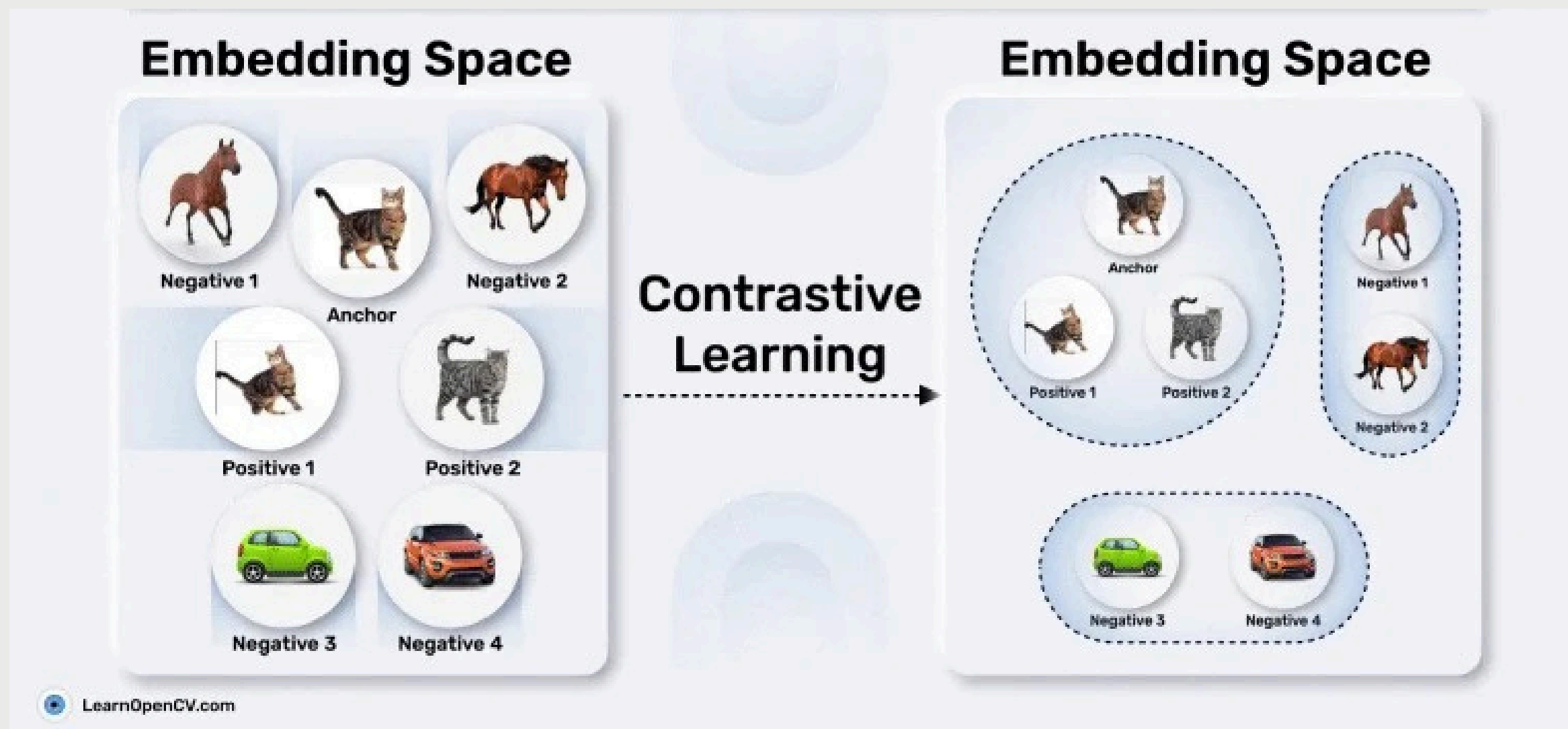


What is Contrastive learning?

Contrastive learning is a machine learning technique that teaches models to distinguish between similar and dissimilar samples. By contrasting different samples, models can learn to identify common attributes and distinguish between classes [1].



Examples

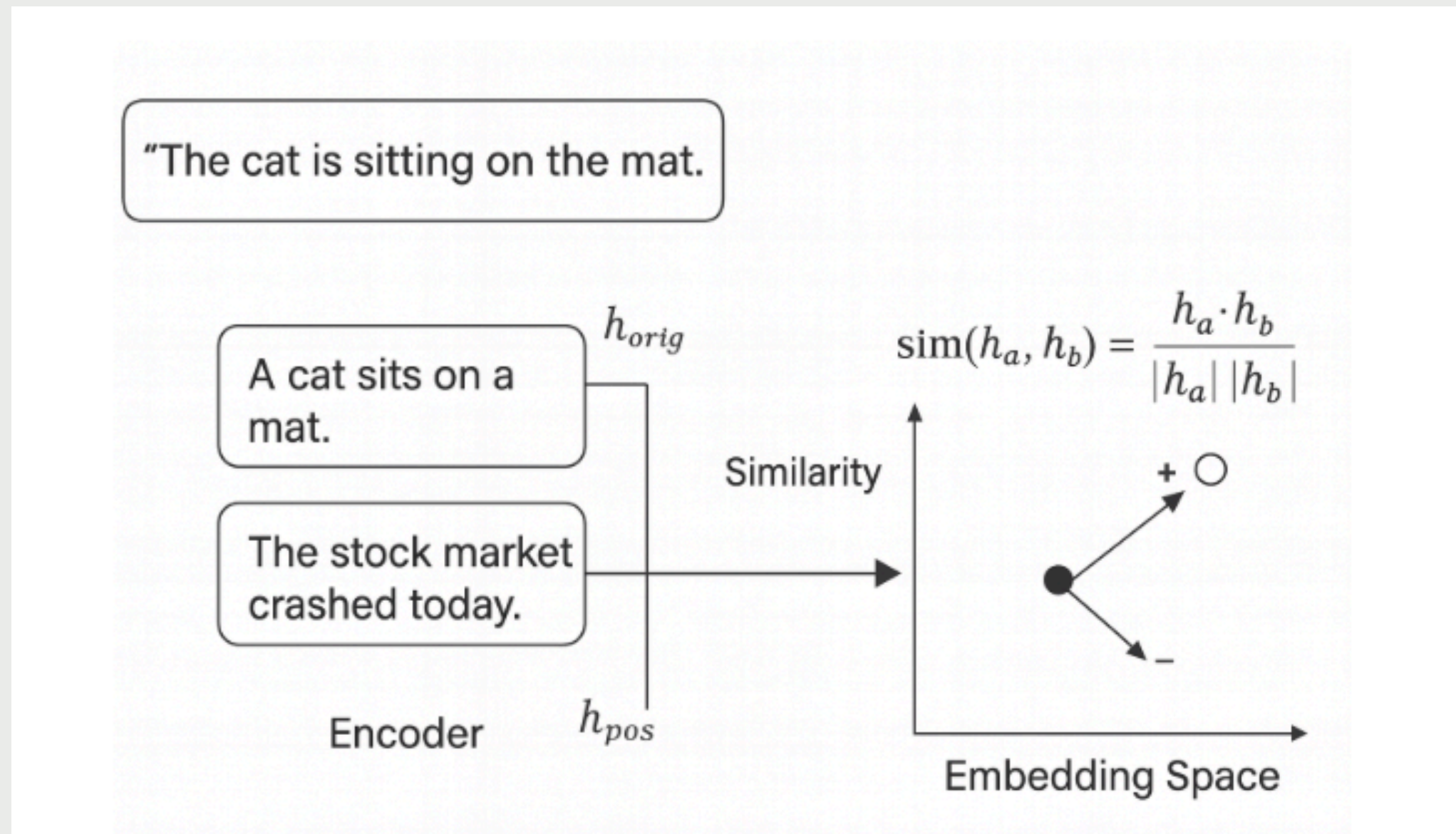


Anchor: the input image (photo of a cat)

Positive Pair: Semantically similar image as the original image (augmented view of the same cat or another different photo from the same class to which the cat belongs)

Negative Pair: Semantically dissimilar image (photo of a car or horse)

Examples



Why do we need Contrastive learning?

- Labeled data is expensive and time-consuming to collect
- e.g. biomedical imaging, where expert doctors are required to annotate the data.
- Unlabeled data is abundant (images, videos, audio, text available at large scale)
- Models need robust representations that work across many conditions - generalisation
- Supervised learning struggles with variations in Lighting, angles and noisy images.
- Real world examples:
 - Face recognition
 - Self-driving cars
 - Document OCR

Learning models commonly used

Self supervised

→ most common way of contrastive learning. It tries to minimize the distance between the anchor and positive samples. Without any labels.

Supervised

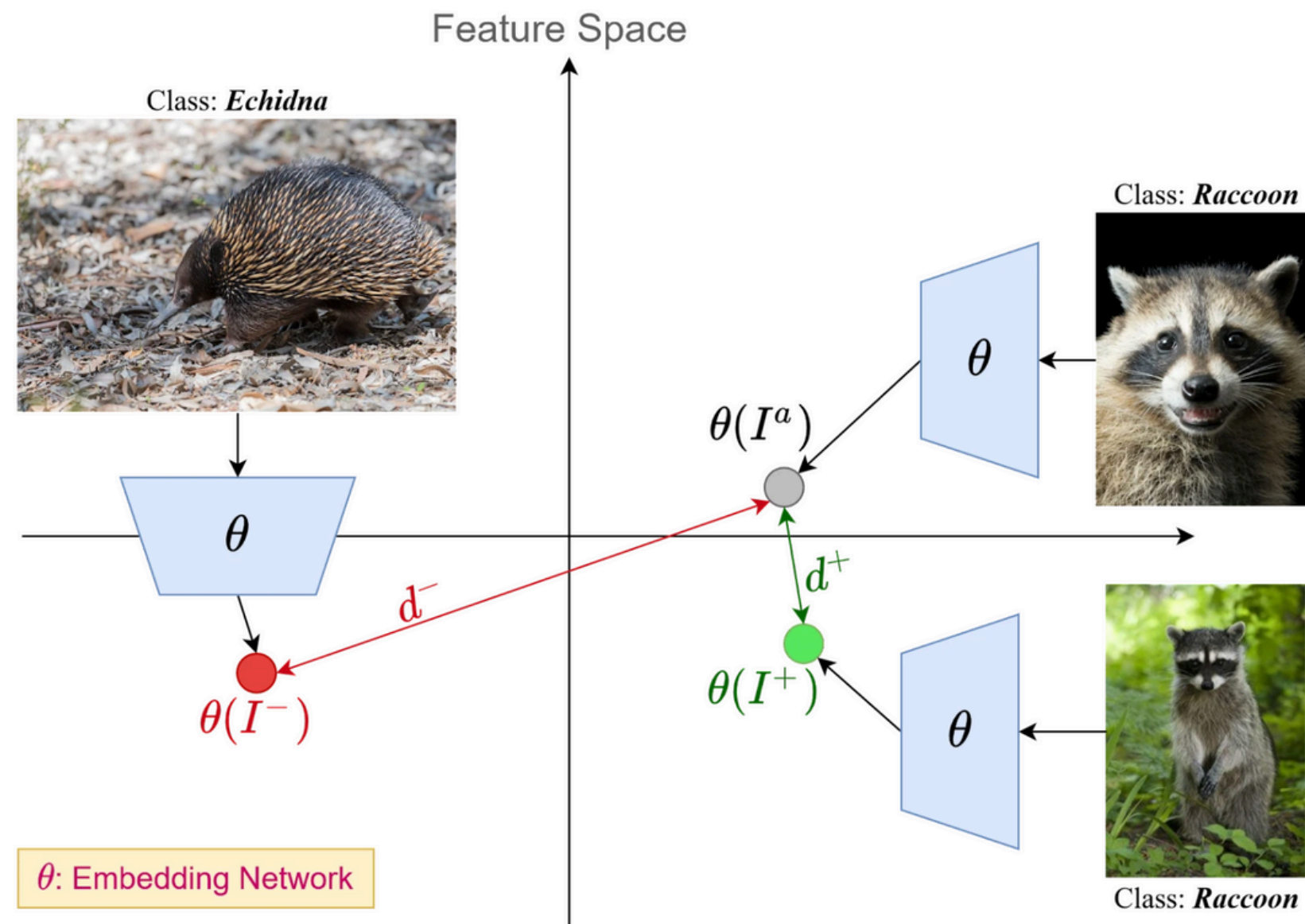
→ It uses labeled information. All samples with same label are positive pairs and all the different labels are negative pairs

Semi Supervised

→ Mix of labeled and unlabeled data. Labeled data guides the contrastive relationships and unlabeled data provides additional learning signal



Self Supervised learning

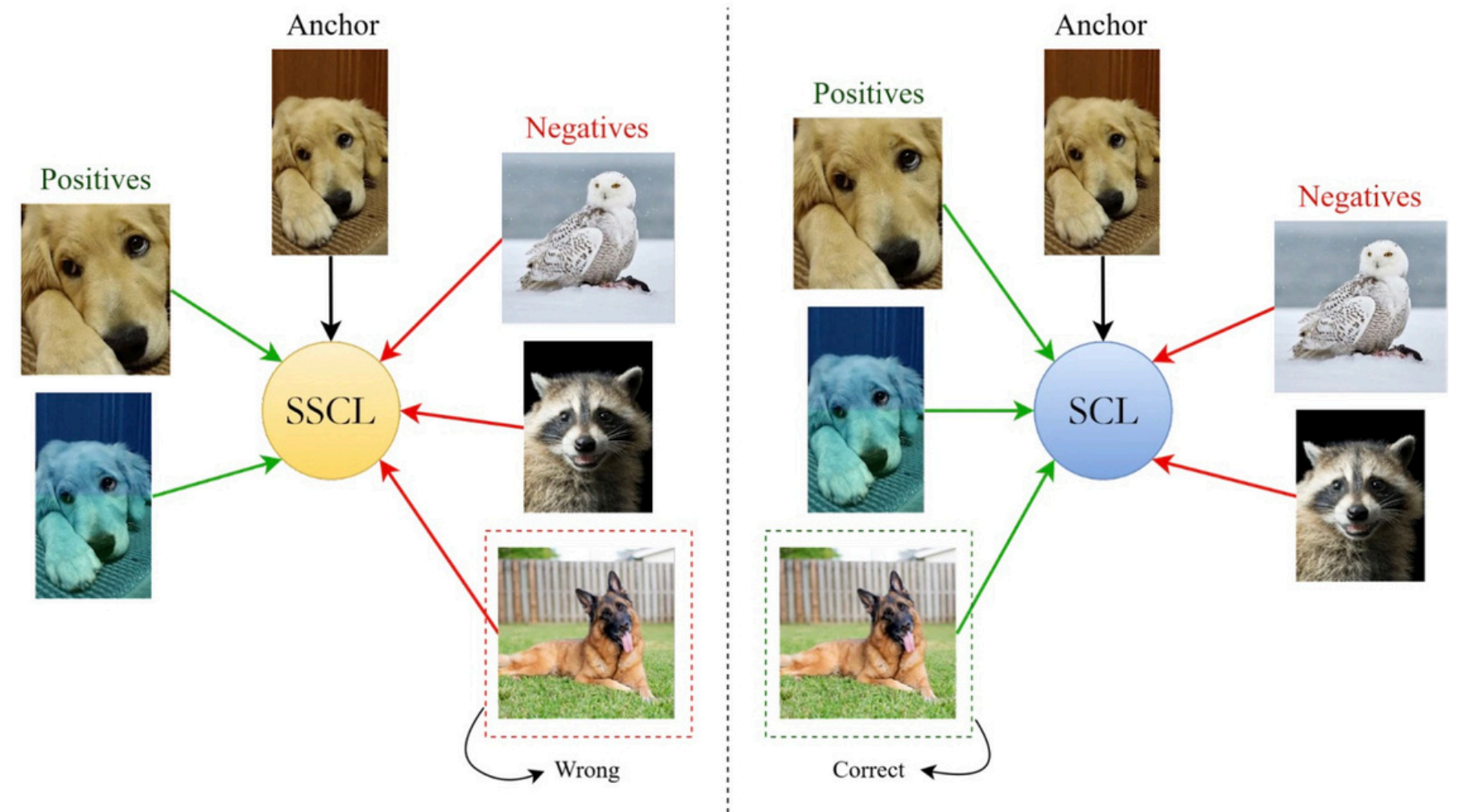


The SSL model tries to minimize the distance between the anchor and positive samples, i.e., the samples belonging to the same distribution, in the latent space, and at the same time maximize the distance between the anchor and the negative sample

The goal of the contrastive learning model (denotes by "theta" in the example above) is to minimize the distance " d^+ " and maximize the distance " d^- ."

- Positive samples are created by augmenting the anchor image (e.g., crop, color jitter, rotation).
- Negative samples are randomly chosen from other images in the minibatch.
- This setup is highly scalable, enabling pre-training on large unlabeled datasets.
- Helps models learn strong feature representations without requiring labels.

Supervised CL vs. Self Supervised CL



One of the major problems faced in Self supervised learning is that only the augmented images of the anchor are considered positive and rest are all considered negative even if they are a part of the same class.

Data Augmentation

Synonym replacement

- Replace words with synonyms:
- The cat sat on the mat → The cat sat on the rug
- Same meaning → good positive pair.
- Mask (hide) a word
- e.g. "The [MASK] crossed the street."

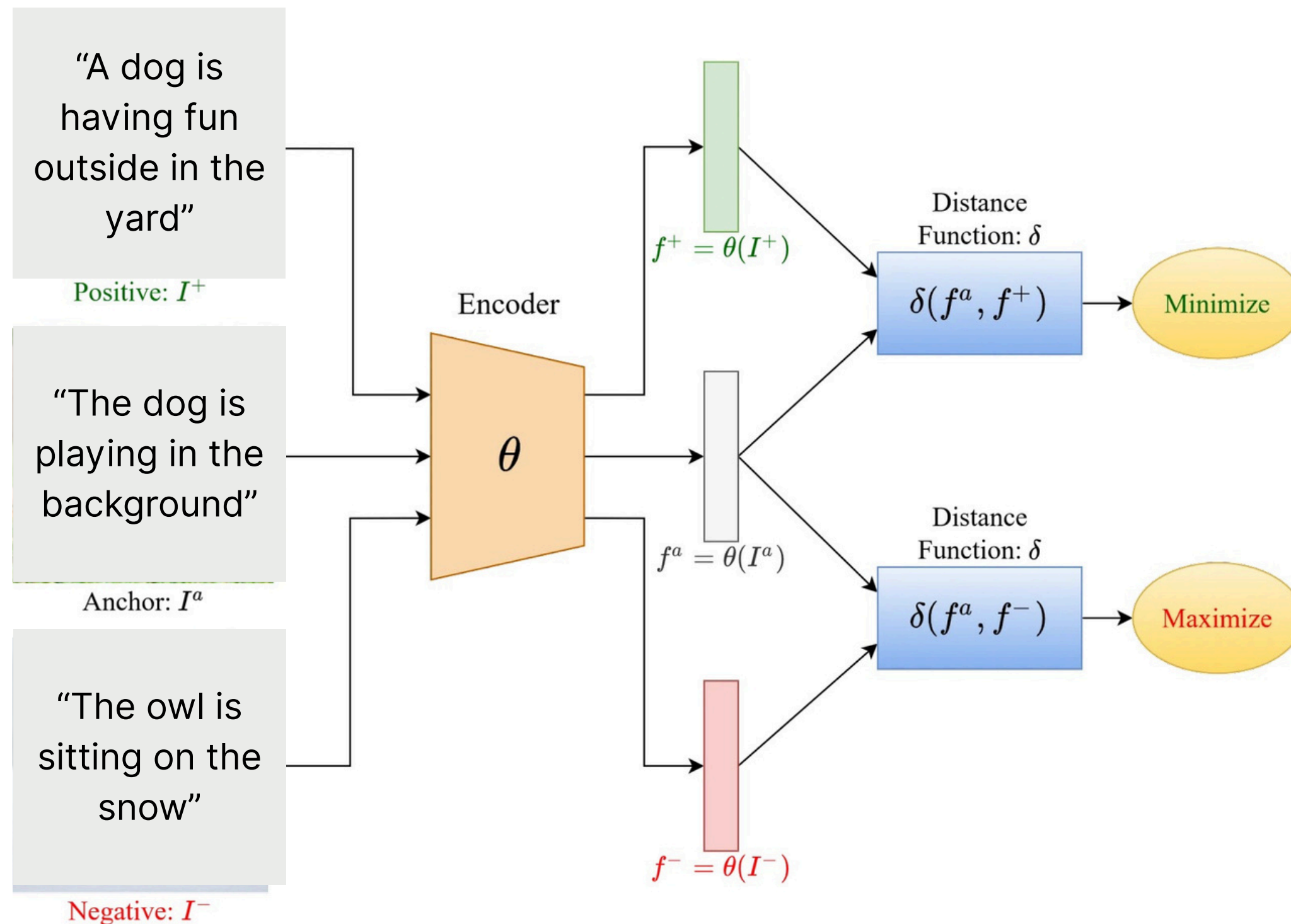
Back translation

- Translate text into another language and back:
 - English → French → English
 - Produces a new sentence with the same meaning but different wording.
- Example:
- "The election results were surprising." → "The outcome of the election was unexpected."

Insertion or deletion

- Delete or insert filler words:
- "I really think this is important" → "I think this is important"
- Maintains core meaning.

Encoder



The encoder network is typically a deep neural network.

What the Encoder does?

Encode the text into embeddings

- $f^a = \theta(\text{anchor sentence})$
- $f^+ = \theta(\text{positive sentence})$
- $f^- = \theta(\text{negative sentence})$

The encoder makes the positive embeddings close and pushes the negative embeddings far apart.

Loss Functions

Loss Function	Formula	Goal / Intuition	Pros	Cons
InfoNCE Loss	$\mathcal{L} = -\log \frac{e^{\text{sim}(h, h^+)/\tau}}{\sum_j e^{\text{sim}(h, h_j)/\tau}}$	Maximize similarity to positives, minimize against all negatives	Works well with large batches; strong representations	Needs many negatives; compute-heavy
NT-Xent Loss	Same as InfoNCE but with normalized embeddings	Scaling similarities uniformly	Stable optimization; common in SSL	Similar drawbacks as InfoNCE
Triplet Loss	$\max(0, d(a, p) - d(a, n) + m)$	Enforce anchor-positive closer than anchor-negative by margin	Intuitive; widely used	Hard negative mining needed
Margin Ranking Loss	$\max(0, m - \text{sim}(h, h^+) + \text{sim}(h, h^-))$	Rank positives above negatives	Good for retrieval tasks	Requires careful margin choice
Cosine Embedding Loss	$\begin{cases} 1 - \text{sim}(h_1, h_2), & y = 1 \\ \max(0, \text{sim}(h_1, h_2) - m), & y = -1 \end{cases}$	Encourage similar pairs to have high cosine sim	Simple and effective	Needs labeled similar/dissimilar pairs

Our Project

Fake News Detection

Testing multiple models to monitor performance of each on semantically different languages - English and Urdu.

We will compare the performance of each Fake News Detection on their respective languages and the comparative language. A multilingual model approach will be utilized to understand bottlenecks and how each performs in their respective tasks in comparison.

Relevance of Contrastive Learning

Contrastive learning explicitly aligns the semantic meaning of news across languages (like English/Urdu) by pulling similar claims together in embedding space, regardless of syntax. This creates a more robust, language-agnostic representation that is less sensitive to class imbalance, as it learns from data relationships rather than just labeled examples.

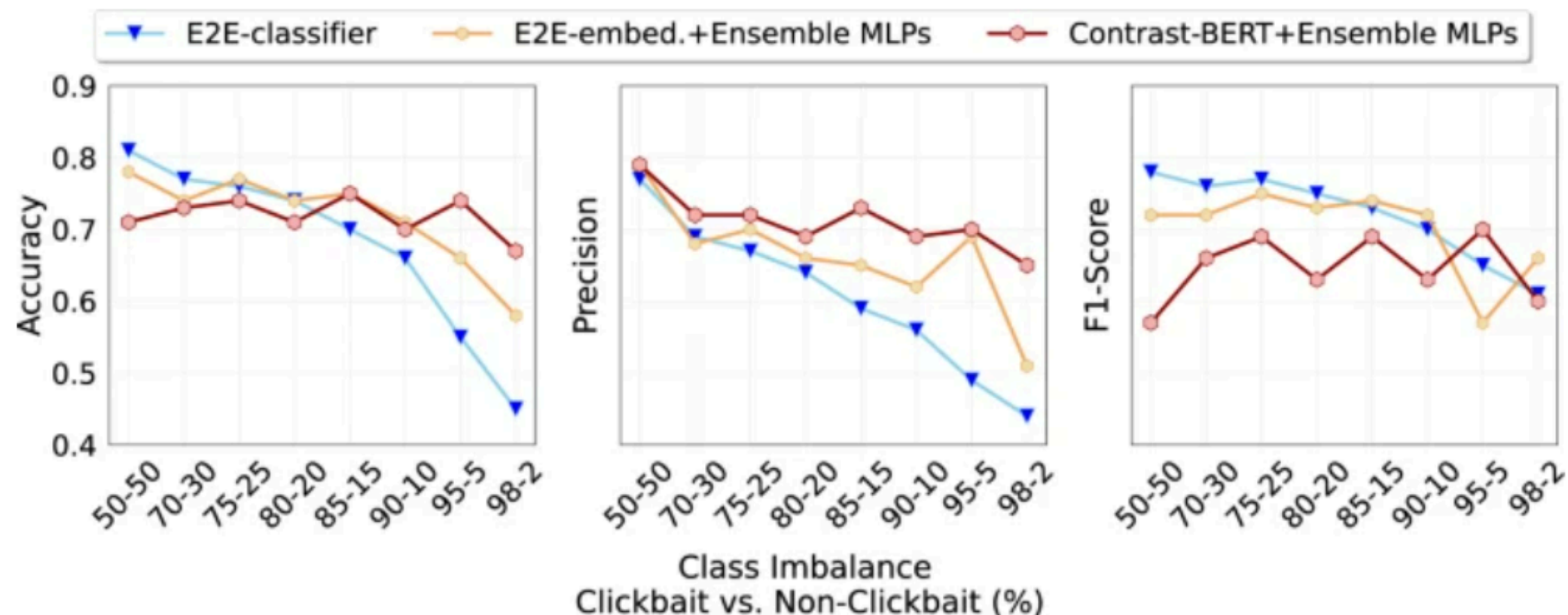
Datasets - Low Resource Languages

Dataset concerns

- Imbalanced
 - class imbalance
 - size imbalance
- Synthetically generated
- Lack of evidence to fact check claims

Benefit of Contrastive Learning

- Adapts to context, not syntax - contextual embeddings
- Preserves both syntactic (grammatical structure) and semantic (meaning) information
- Integration of external knowledge



Bhattacharjee et al. (2022) - "Text Transformations in Contrastive Self-Supervised Learning"

- Core Challenge: Unlike images, text augmentations risk semantic shift - modifying even a single word can change meaning, making positive pair construction fundamentally harder than in computer vision
- Key Challenge in Text CL:
 - Unlike images (crop, rotate, flip), text transformations are non-trivial
- Three Categories of Text Transformations:
 - Input-Space Transformations: Direct token-level operations
 - Word/span deletion, token reordering, synonym substitution (EDA methods)
 - Span sampling: selecting adjacent/overlapping segments from same document as positive pairs (DeCLUTR)
 - Risk: May inadvertently alter semantics and violate positive pair assumptions
- Latent-Space Transformations: Transformations via learned representations
 - Back-translation: translate to another language and back (preserves semantics)
 - Language model-based word replacement using embedding nearest neighbors
 - Contiguous span masking/cropping through language models

Bhattacharjee et al. (2022) - "Text Transformations in Contrastive Self-Supervised Learning"

- Architecture-Based Methods: Network-driven transformations
 - Different dropout masks on same input → different embeddings (SimCSE approach)
 - Adversarial perturbations in latent space - add noise - embeddings change
 - Using NLI relations: entailment pairs as positives, contradiction/neutral as negatives
- Negative Sampling Strategies:
 - Challenge: Without labels, may accidentally sample semantically similar instances as negatives (false negatives hurt learning)
- Solutions:
 - Increase batch size for more negatives (memory intensive, may decrease performance)
 - Hard negative mining: select negatives semantically close but from different classes
 - Approximate nearest neighbor (ANN) search across entire dataset (not just in-batch)
- Contrastive Loss Function

References

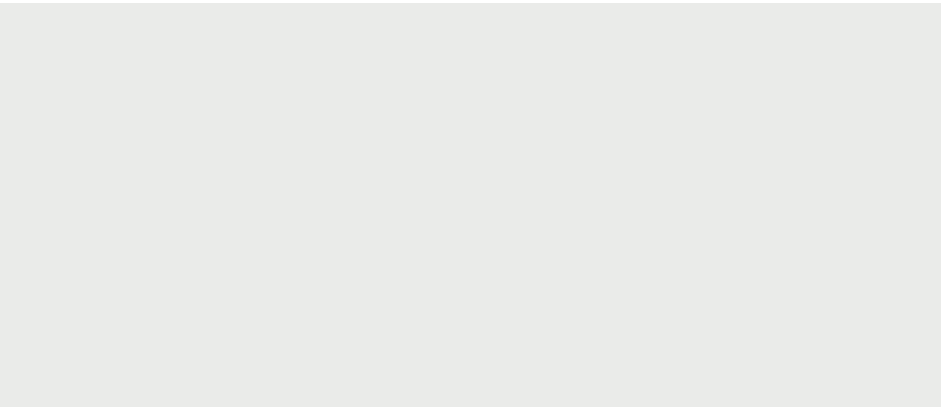
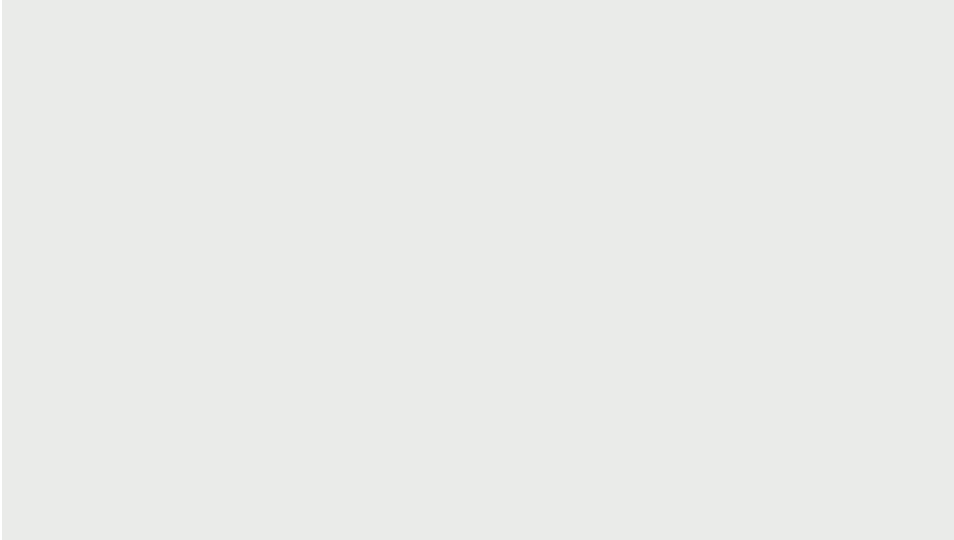
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**THANK
YOU**

21 Nov, 2025

